

SIVAPRIYA GOPI

Minneapolis, MN | ☎ +1763-900-2914 | ✉ gopi0011@umn.edu | [in](#) [Linked In](#) | [GitHub](#) | [Sivapriya Gopi's Portfolio](#)

Education

Master of Science in Computer Science

University of Minnesota, Twin Cities - **GPA: 3.9/4**

Sep 2024 – May 2026

Minnesota, USA

Bachelor of Technology in Computer Science and Engineering

SRM Institute of Science and Technology - **CGPA: 9.57/10**

Aug 2020 – May 2024

Chennai, India

Technical Skills

Programming Languages & Data Analysis: Python, C, C++, Java, JavaScript, SQL, MATLAB, HTML, CSS, VBA, NumPy, Pandas

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Django, React, Spring Boot, REST APIs, Svelte

Financial Modeling & Reporting: Excel (advanced formulas, Pivot Tables, dashboards), Power BI, Power Query, Business Objects

Business Intelligence & Data Pipelines: Tableau, PowerBI, Alteryx, ETL workflows, Power Automate, CI/CD

AI, Machine Learning & Automation: BERT, GPT-4, Llama-3, NLP, Reinforcement Learning, Generative AI (LLMs, RAG), HuggingFace

Cloud, Development & Enterprise Tools: Azure, AWS, Docker, GitHub, Microservices, SDLC, Microsoft 365 (Power Suite, Copilot Studio)

Communication & Collaboration: PowerPoint storytelling, Cross-functional teamwork, Project management, Analytical problem-solving

Professional Experience

Graduate Research Assistant - Department of Neurosurgery

University of Minnesota, Twin Cities

Dec 2024 – Present

Minneapolis, MN

- Engineered **Python & MATLAB applications** to automate ingestion and perform statistical data analysis of multimodal patient datasets (30+ cases), cutting preparation time from hours to less than 20 minutes.
- Designed **Power BI dashboards & Excel reports** that delivered clinician-ready insights, accelerating neurosurgeon decision-making and improving study accuracy.
- Automated repetitive workflows (e.g., **Brain and Motor Control Assessment, TcD, ABPM**), reducing turnaround time by ~40% and minimizing human error.
- Developed a custom **Dual N-Back cognitive testing web app** with synchronized visual/auditory stimuli and real-time scoring, now integrated into CE-STAND clinical trials.

Software Developer Intern – Backend Systems

Softlogic Systems

Jun 2023 – Sep 2023

Chennai, India

- Designed and deployed scalable **Java Spring Boot backends** and **RESTful APIs** for payroll systems serving 5 Healthcare organizations (250+ employees each).
- Implemented **OAuth 2.0 authentication**, securing sensitive financial and HR data.
- Gained hands-on experience designing queries and schemas in **PostgreSQL and SQL Server**, optimizing performance and caching strategies to reduce response time by **30%** and enhance reporting speed.
- Leveraged **Chroma and Pgvector** to enhance data retrieval efficiency and scalability, improving reporting accuracy by **20%** and enabling data-driven workforce operations across industries.

Projects ([GitHub Repositories](#))

Multi-Agent Hint Protocol for Adaptive Trust and Deception in LLMs

Python, GPT-4, Llama-3, Mistral, NLP, Prompt Engineering, Reinforcement Dynamics, Data Visualization

Sep 2025 – Dec 2025

- Built a hint-based, auto-scored multi-agent framework to analyze cooperation, deception, and trust calibration in large language models.
- Implemented prompt-time adaptation using lightweight trust memory and incentive-controlled dynamics for behavioral analysis.

Stock Sentiment Analysis – The Social Media Effect

Python, BERT, NLP, Svelte, Pandas, Data Visualization

Jan 2025 – May 2025

- Analyzed 50K+ tweets and stock prices using **BERT-based NLP** and causality modeling, finding short-term impacts on 11 US stocks.
- Developed an interactive **Svelte dashboard** for data storytelling and visualization of trading trends.

Intruder Escape – AI Pursuit-Evasion Game

Python, Pygame, A* Pathfinding, Uniform Cost Search

Sep 2024 – Dec 2024

- Designed a pursuit-evasion simulation reducing node exploration by **4.4x** with optimized search algorithms.
- Implemented difficulty scaling and dynamic obstacle generation for adaptive AI behavior in real time.

News Headline Classification Model

Python, Scikit-learn, Pandas, NLP, Supervised Machine Learning, Data Visualization

Jan 2024 – May 2024

- Built ML pipeline classifying 25K+ news headlines across categories with **92% accuracy**.
- Applied tokenization, vectorization, and multi-class classification models.